produce, when they collide and coalesce to form a single hole? Hawking had been vaguely aware for some time that the single final hole would have to be larger, in some sense, than the "sum" of the two original holes, but in what sense, and what could that tell him about the amount of gravitational radiation produced?

Then, as he was preparing for bed, it had hit him. Suddenly, a series of mental pictures and diagrams had coalesced in his mind to produce the Idea: It was the area of the hole's horizon that would be larger. He was sure of it; the pictures and diagrams had coalesced into an unequivocal, mathematical proof. No matter what the masses of the two original holes might be (the same or very different), and no matter how the holes might spin (in the same direction or opposite or not at all), and no matter how the holes might collide (head-on or at a glancing angle), *the area of the final hole's horizon must always be larger than the sum of the areas of the original holes' horizons.* So what? So a lot, Hawking realized as his mind roamed over the ramifications of this *area-increase theorem*

First of all, in order for the final hole's horizon to have a large area, the final hole must have a large mass (or equivalently a large energy), which means that not too much energy could have been ejected, as gravitational radiation. But "not too much" was still quite a bit. By combining his new area-increase theorem with an equation that describes the mass of a black hole in terms of its surface area and spin, Hawking deduced that as much as 50 percent of the mass of the two original holes could be converted to gravitational-wave energy, leaving as little as 50 percent behind in the mass of the final hole.[1]

There were other ramifications Hawking realized in the months that followed his sleepless November night. Most important, perhaps, was a new answer to the question of how to *define* the concept of a hole's horizon when the hole is "dynamical," that is, when it is vibrating wildly (as it must during collisions), or when it is growing rapidly (as it will when it is first being created by an imploding star).

Precise and fruitful definitions are essential to physics research. Only after Hermann Minkowski had *defined* the absolute interval between two events (Box 2.1) could he deduce that, although space and time are "relative," they are unified into an "absolute" spacetime. Only after Einstein had *defined* the trajectories of freely falling particles to be straight lines (Figure 2.2) could he deduce that spacetime is curved (Figure 2.5), and